

Ferroelectric Recording for Ultra-High Areal Density Storage

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The scaling of heat-assisted magnetic recording (HAMR) beyond ~ 10 TB/disk is primarily constrained by the minimum magnetic grain size required to preserve both signal-to-noise ratio and thermal stability. Ferroelectric media provides a potential alternative by storing information in stable polarization states. Due to intrinsically narrow and low-energy domain walls, ferroelectric domains can, in principle, be scaled to near-lattice-constant dimensions. In this work, we investigate ferroelectric domain stability at nanometer length scales and demonstrate the recording of polarization patterns in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ (HZO) thin films using phase-field simulations. We show that ferroelectric domains can remain stable at smaller lateral dimensions than magnetic bits under comparable thermal stability constraints. These results suggest that ferroelectric recording in continuous films may provide a viable route beyond HAMR toward next-generation storage densities.

Index Terms—Ferroelectric recording, HZO, phase-field modeling, HDDs

I. INTRODUCTION

The rapid growth in data storage demand, particularly from AI-enabled applications, has driven continued scaling of magnetic recording technologies. Heat-assisted magnetic recording (HAMR) has emerged as a leading approach to extend areal density by enabling the use of high-anisotropy media through localized laser heating during the write process.

The ultimate scalability of HAMR is primarily constrained by requirements on the media microstructure. At areal densities beyond ~ 10 – 12 TB/disk, recording systems require track pitches (TP) below ~ 15 nm and bit lengths (BL) below ~ 10 nm. This requires reducing the media grain size below ~ 5 nm while maintaining sufficient grain volume to ensure thermal stability ($K_u V / k_B T$), preserving strong crystalline anisotropy, and mitigating finite-size effects. In addition, grain boundary dimensions must be simultaneously scaled to retain a high packing fraction without degrading magnetic isolation.

Alternatively, ferroelectric materials exhibit a natural separation between stability and scalability. The thermal stability of stored information is governed by the bulk Landau free-energy barrier, which scales with the domain volume ($\Delta U \propto L^3$) and ensures robust polarization retention. In contrast, the domain wall energy density in ferroelectric materials is relatively small; domain walls can be formed at low energetic cost, enabling the creation of densely packed nanoscale domains [1]. This decoupling allows ferroelectric systems to support much smaller feature sizes than magnetic media. As a result, ferroelectric materials can provide a favorable pathway for scaling to ultrahigh areal densities, provided that electrostatic effects are properly managed [2].

II. FERROELECTRIC DOMAIN STABILITY

The total free energy of a ferroelectric system is given by

$$F = \int (f_{\text{bulk}} + f_{\text{grad}} + f_{\text{elec}} + f_{\text{elas}}) dV, \quad (1)$$

where f_{bulk} is the Landau double-well potential, $f_{\text{grad}} = \kappa (\nabla P)^2$ is the gradient energy that penalizes spatial variations

in polarization, f_{elec} accounts for depolarization fields, and f_{elas} captures elastic coupling.

The bulk term sets the intrinsic energy barrier and governs thermal stability, while the gradient term determines the domain wall energy and width. The electrostatic term becomes critical at nanoscale dimensions, where incomplete screening can destabilize the polarization.

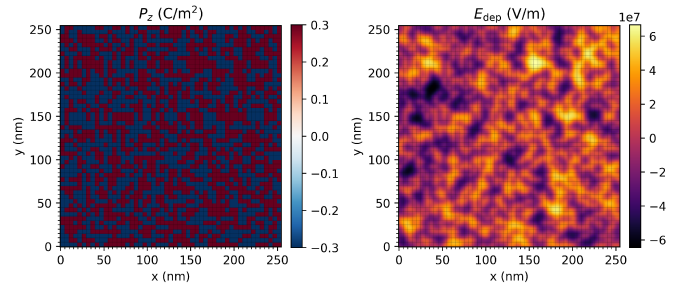


Fig. 1. (a) Checkerboard polarization pattern (P_z) and (b) corresponding depolarization field (E_{dep}) computed with grounded bottom-electrode and 80% top-surface charge compensation ratio ($\eta_{\text{top}} = 0.8$).

For an ultrafine HZO domain with dimensions of $5 \times 5 \times 10$ nm³, the bulk energy barrier is estimated to be $\Delta U \approx 1024 k_B T$, indicating strong thermal stability at room temperature. The gradient energy contribution is comparatively small owing to the low gradient coefficient ($\kappa \sim 10^{-12} \text{ J m}^3/\text{C}^2$) [1].

In contrast, depolarization fields can significantly reduce stability when bound surface charges are not properly compensated. As shown in Fig. 1b, the depolarization field can become locally strong and lowering the energy barrier for thermally driven polarization reversal. This effect can significantly impact data retention in ferroelectric recording.

Fig. 2 illustrates how the energy barrier distribution changes as a function of the top-surface charge compensation ratio (η_{top}). For each value of η_{top} , 200 random checkerboard configurations similar to the pattern shown in Fig. 1a are generated and their energy barriers computed. As η_{top} decreases from unity, incomplete screening strengthens the depo-

larization field, broadening the barrier distribution and shifting it toward lower values. This indicates an increased probability of thermally driven polarization reversal in domains with poor surface compensation. The cumulative distribution function (CDF) is shown in Fig. 2b. The probability of a domain having an energy barrier below a given threshold, $P(\Delta U/k_B T \leq x)$, increases substantially as η_{top} decreases, indicating that a larger fraction of domains become thermally unstable. The fixed point at which all CDF curves intersect corresponds to the bulk Landau free-energy barrier. At this point the average depolarization field contribution is zero, and the barrier distribution becomes independent of the surface charge compensation.

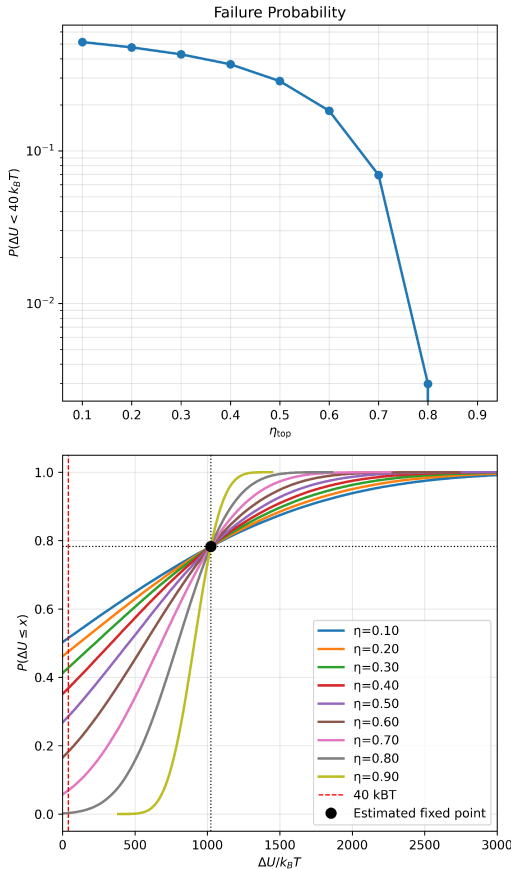


Fig. 2. (a) probability of having thermally unstable domain ($\Delta U < 40 k_B T$) and (b) cumulative distribution function for varying top-surface charge compensation ratios (η_{top}) with grounded bottom-electrode.

For surface charge compensation above $\sim 80\%$, the depolarization effect becomes minimal and data retention improves significantly (Fig. 2a). In addition, sensitivity to surface charge compensation can be tuned by the substrate strain and favorable retention can be achieved even at lower η .

III. FERROELECTRIC RECORDING ON HZO

The polarization dynamics can be modeled by the time-dependent Landau–Ginzburg–Devonshire (TDLGD) equation:

$$\frac{\partial P}{\partial t} = -\Gamma \frac{\delta F}{\delta P}, \quad (2)$$

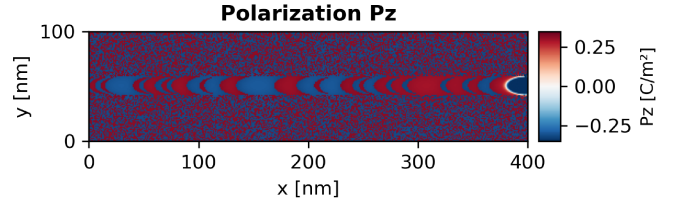


Fig. 3. PRBS recording track written on HZO by a moving Gaussian electric field ($\sigma_x = 10 \text{ nm}$, $\sigma_y = 5 \text{ nm}$, $E_0 = 10^8 \text{ V/m}$) at velocity $v = 20 \text{ m/s}$. Compressive misfit strain ($u_m = -0.01$) is imposed at the substrate interface with a stress-free top surface. The bottom electrode is grounded; the top surface has 80% surface charge compensation.

where Γ is the kinetic coefficient and F is the total free energy defined in Eq. (1). This framework captures domain nucleation, domain wall motion, depolarization effects, and the evolution of polarization under applied electric fields.

In Eq. (2), the depolarization and externally applied fields enter through the electrostatic contribution to F . In this work, the Landau coefficients are renormalized to incorporate electrostrictive coupling under the assumed mechanical boundary conditions: a compressive misfit strain $u_m = -0.01$ imposed by the substrate at the bottom interface and a stress-free top surface ($\sigma_{33} = 0$). The recording medium consists of a 10 nm thick HZO film, discretized with a mesh size of $dx = dy = 1 \text{ nm}$. The polarization is assumed to be uniform along the out-of-plane direction.

Using phase-field model, we simulate the recording process under a moving localized electric field that mimics a write head. A Gaussian field profile ($\sigma_x = 10 \text{ nm}$, $\sigma_y = 5 \text{ nm}$, $E_0 = 10^8 \text{ V/m}$) moves with velocity $v = 20 \text{ m/s}$ and nucleates polarization domains. Stable track-like recording patterns persist after field removal (Fig. 3).

Unlike magnetic media, where grain size fundamentally limits bit stability, ferroelectric domains can be formed at nano scales owing to low domain wall energy and the absence of exchange-coupling constraints. Recorded bits smaller than $\sim 5 \text{ nm}$ and tracks narrower than $\sim 10 \text{ nm}$ are achievable, corresponding to areal densities exceeding 12 Tb/in^2 .

IV. CONCLUSION

We have presented a phase-field study of ferroelectric recording in HZO thin films. The analysis shows that ferroelectric systems exhibit fundamentally more favorable energy scaling than magnetic media. Our results indicate that ferroelectric recording provides a promising pathway to surpass HAMR scaling limits and enable next-generation high-density storage technologies.

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